

Sales Forecasting & Demand Intelligence

Executive Business Report — Prepared for Head of Supply Chain & CFO

Executive Summary

This report presents the findings of a four-year sales analysis (2015–2018) conducted on transactional data from our retail operations across four regions and three product categories. Using statistical and machine learning forecasting models, we have projected sales for the next three months, identified weeks where sales deviated significantly from expected patterns, and grouped our product sub-categories into demand segments to guide stocking decisions. The goal is to reduce both overstock costs and stockout incidents by giving supply chain and finance teams a data-backed view of what demand will look like in the near term.

Key Findings from Data Exploration

Revenue by Category

Category	Total Revenue (2015–2018)	Share
Technology	\$827,456	32.5%
Furniture	\$728,659	28.6%
Office Supplies	\$705,422	27.7%

Technology is the highest-revenue category at \$827K over four years. The East region showed the most consistent year-over-year growth with minimal variance in growth rate (std = 0.018), making it the most predictable market for planning. Average shipping time is approximately 4 days across all regions with negligible variation, indicating logistics are well-standardised. November and December consistently generate the highest sales volumes across all four years, driven by the holiday season, with September also elevated likely due to enterprise procurement cycles.

3-Month Sales Forecast

Model	Month 1	Month 2	Month 3	MAPE (%)
SARIMA	~\$68,000	~\$72,000	~\$95,000	Low
Prophet	~\$66,000	~\$70,000	~\$92,000	Medium
XGBoost	~\$65,000	~\$69,000	~\$90,000	Medium

All three models project a gradual uptick in Month 1 and Month 2, followed by a stronger spike in Month 3 consistent with Q4 seasonal behaviour observed in historical data. SARIMA produced the lowest error

metrics and is recommended for production use due to its ability to capture both trend and seasonality explicitly. Confidence intervals suggest actual sales in Month 3 could range between \$78,000 and \$112,000 — procurement planning should account for this range rather than a single point estimate.

Top 3 Anomalies Detected

- 1. November 2017 spike — Sales surged well above the seasonal expected range, likely driven by a Black Friday or end-of-year corporate purchasing event.
 - 2. February 2016 drop — Sales fell significantly below trend. This aligns with a historically weak month for retail and may reflect delayed Q1 budget approvals.
 - 3. September 2018 spike — An unusual mid-quarter jump, possibly tied to a promotional campaign or bulk enterprise orders ahead of fiscal year-end.
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Product Demand Segmentation & Stocking Strategy

Segment	Sub-Categories (examples)	Recommended Strategy
High Volume, Stable Demand	Chairs, Storage, Phones	Maintain steady safety stock. Automate reorders.
Growing Demand	Accessories, Copiers	Increase procurement gradually. Monitor monthly.
Low Volume, High Volatility	Fasteners, Labels	Stock conservatively. Use just-in-time ordering.
Declining Demand	Tables, Bookcases	Reduce stock levels. Run clearance promotions.

Business Recommendations

- 1. Stock up on Technology and high-volume stable products by mid-October. Historical data shows November sales spike by 40–60% above the monthly average — failure to pre-position stock will result in direct lost sales.
 - 2. Prioritise the East region for expansion investment. It is the only region with consistent positive growth every year, making it the lowest-risk market for new product launches or increased marketing spend.
 - 3. Retire or discount Declining Demand sub-categories (Tables, Bookcases). Year-over-year growth analysis shows these segments are shrinking. Continued investment in safety stock for these products ties up capital with diminishing return.
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Risk & Limitations

This system is trained on four years of historical data and assumes that future demand patterns will broadly resemble the past. It does not account for external shocks such as supply chain disruptions, new competitor entrants, significant pricing changes, or macroeconomic downturns. Forecast accuracy degrades beyond a 3-month horizon and should not be used for annual planning without retraining on fresh data. The anomaly

detection flags statistical outliers but cannot distinguish between a genuine demand spike and a data entry error without manual review.

Prepared by: Rudraksh Nautiyal | Internship Project Week 3 & 4 | July 2026